Multi-Word Expressions (MWE), Linking Verbs & Light Verbs Constructions (LVC)

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What are Multi-Words Expressions (MWE)?

Multi-word expressions are linguistic objects formed by two or more words that behave like a "unit" semantically. Their real meaning comes from considering the words together. Both nouns and verbs can come as MWEs (e.g., the noun "United States of America" or the verb "be in charge of" something, to "look tired").

Improving SVO Output Semantic Readability of Multi-Word Expressions (MWE)

To improve the readability of multi-word expressions (MWE), the SVO algorithm has several ways of dealing with MWEs, in order to capture the full meaning of these linguistics expressions.

1. For **nouns** (e.g., United States of America), the SVO algorithm relies on the **NER tags** "**entitymentions**" ("text": "United States of America"), rather than the separate values in "word" or "originalText" ("United," "States," "of," "America") exported to the JASON file.
2. For **verbs**, in the absence of NER values which apply to nouns only, the SVO algorithm relies on functions to deal with **linking verbs** and **light verbs** (LV or LVC, Light Verb Construction), special cases of MWEs.

What are Linking Verbs?

According to *The Oxford Learners' Dictionary* a linking verb is "a verb such as *be* or *become* that connects a subject with the adjective or noun (called the complement) that describes it." According to the *Merriam-Webster Dictionary*, a linking verb is "an intransitive verb that connects a subject with a word or words in the predicate. The words 'look' in 'you look tired' and 'are' in 'my favourite fruits are apples and oranges' are linking verbs." The *Cambridge*

Dictionary defines a linking verb as "a verb that connects the qualities of an object or person to that object or person." Thus, in the sentence "My bags weigh 45 kg," "weigh" is a linking verb.

Examples of Linking Verbs

Common linking verbs include "be," "become," "seem," "appear," "look," "feel," "smell," "sound," "taste," "stay," "grow," "turn," "prove," and "remain."

What are Light Verbs?

In linguistics, a **light verb is a verb that has little semantic content of its own** and forms a predicate with some additional expression, which is usually a noun. Common verbs in English that can function as light verbs are: *be, do, give, have, make, get,* and *take*.

Light verbs are a subclass of multiple word expressions (MWE).

Most light verb constructions in English include a noun and are sometimes called stretched verbs.

Examples of LVCs

Some light verb constructions also include a preposition, e.g.

be happy for
be in charge of
be responsible for
be sick of
do the review of
do the cleaning
get support for
get support from
have a nap
have a smoke
have a conversation
give a presentation
give a hard time
give a hug
make a mistake
make a request
take a break
take a nap
take a picture
take a shower
take advantage of
take care of

take charge of
take into consideration

Editing LVCs & Linking files to improve CoreNLP output semantically

To improve the semantic readability of SVO from Stanford CoreNLP, the NLP Suite provides three txt files that the user can edit for Linking verbs and LVCs. The files are stored in NLP-Suite\lib\CoreNLP_enhanced_dependencies.

These files provide basic JASON templates to process special verb MWE.

Notice: while manually editing the Jason txt files, pay attention to the commas between each item in the dictionary and the indentation of "{}" and "[]" (indent = 4 when the json was created by code).

Verb + preposition

file: NLP/lib/CoreNLP_enhanced_dependencies /verb_prep_json.txt

In the csv output of certain corpus, you may find that a verb's object is not extracted, and there's a preposition between the verb and the object.

For example, in the sentence "Then she put on her lynx jacket, almost stepped over me with a conventional excuse in pure Latin American Spanish, left without even saying good-bye or at least thanking me for all I had done to make our night together a happy one, and disappeared into the sun of today in the Amazon jungle of New York." (Strange Pilgrims, Gabriel Garcia Marquez, 1992), "me" is the object of the verb "step", and there's a preposition "over" between them. However, in the output, there's no "step" in the verb column.

S	V	O/A
she	put on	jacket
Someone?	saying	bye or thanki
Someone?	thanking	me
I	done	
I	make	night
I	disappeared	

In this case, the verb-preposition collocation "step over" can be added in to the json file to help the algorithm extract certain verbs and objects:

1. Open the file, and find if the object ("over" in this case) in the keys of the json



2. If the preposition is in the keys, add “step” in the list, which is the content to each key.
3. If it’s not found, added the key and content (an empty list). It would be better to added the key at the correct position according to alphabetical order.



4. add the verb.



```
        "play",
        "pride",
        "work"
    ],
    "over": [
        "step"
    ],
    "to": [
        "add",
        "apologize",
        "apply",
        "belong",
        "compare",
        "complain",
        "consent",
        "contribute",
```

Verb + pseudo object + real object

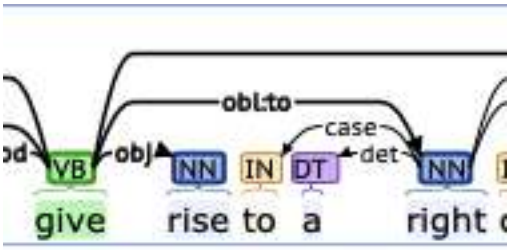
file: NLP/lib/CoreNLP_enhanced_dependencies/verb_obj_obl_json.txt

In this sentence "The provisions of this MOU will not give rise to a right of any person to impede the execution of a request.", "give rise to" should be considered as a **light verb construction** that plays the role of a verb, while "right" is its object. However, in the output, that light verb construction is not recognized:

V	O/A
give	rise

In this case, the light verb "give rise to" can be added in to the json file to help the algorithm extract certain verbs and objects:
1. Go to this website (https://corenlp.run/ or https://stanfordnlp.github.io/CoreNLP/demo.html and click the demo link), then past the sentence containing that light verb and run:

2. Check the dependencies:



3. Open the txt file and find if the verb ("give" in this case) in the keys of the json. If it is, add an empty dictionary to its content, which is a list. If not, add the key and an empty list as its content.

```
"give": [
    {
        |
    },
    {
        "obj": "toss",
        "nmod": "about",
        "downwards": "obj"
    },
```

4. Check the dependencies:

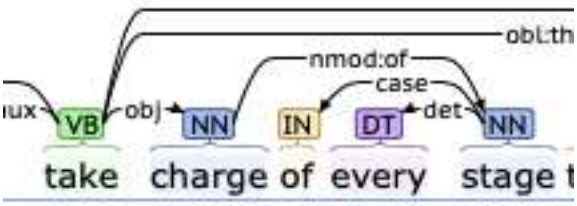
   i. since the dep of “rise” is obj and “give” is its syntactical head, add “obj”: “rise” to the dictionary

```
"give": [
    {
        "obj": "rise"
    },
```

   ii. Find the dep of the object (“obj:to”), then split the deprel at “:”, use the first part as the key and second part as content, and add it to the dictionary (“obl:to”)

```
"give": [
    {
        "obj": "rise",
        "obl": "to"
    },
```

   iii. In some cases, the verb is not the direct syntactical head of the object, such as in the LVC “take charge of”:



   In this case, the syntactical head of the real object “stage” is the pseudo object “charge”. If the real object is governed by a token whose syntactical head is the verb, add “downwards” as a key, and that token’s dep as the content (here the dep of “charge” is “obj”)

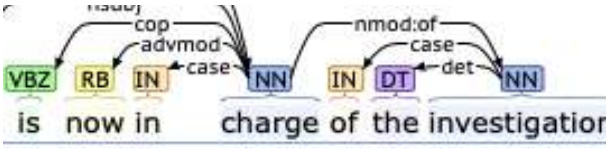
```
"take": [
    {
        "obj": "care",
        "obl": "of"
    },
    {
        "obj": "advantage",
        "obl": "of"
    },
    {
        "obj": "charge",
        "nmod": "of",
        "downwards": "obj"
    }
],
```

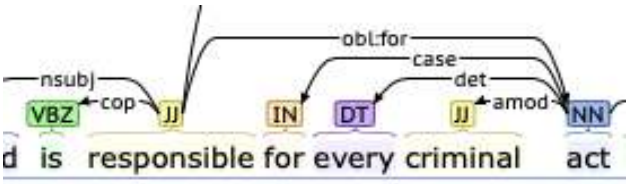
LVC that start with a link verb

file: NLP/lib/CoreNLP_enhanced_dependencies/link_verb_LVC_json.txt

Some LVCs start with a link verb, such as “be responsible for”, “be in charge of”. To recognize this LVCs:

1. Run Stanford CoreNLP online demo to visualize the dependency:





2. Find the “ROOT” within this LVC, which is the token that is the syntactical head of all other tokens (except for the preposition) as well as the real object. For instance, in the LVC “in charge of”, “charge” is the syntactical head of “is”, “in”, “investigation”, but not the preposition “of”

3. Add root (“charge”, “responsible”) into the dictionary as a key, with a content of an empty list including an empty dictionary.

```
"charge": [
    {

    }
],
```

4. Add other tokens in the LVC (except for the “preposition”) in to the empty dictionary to the dictionary in the form of token’s lemma : dep:

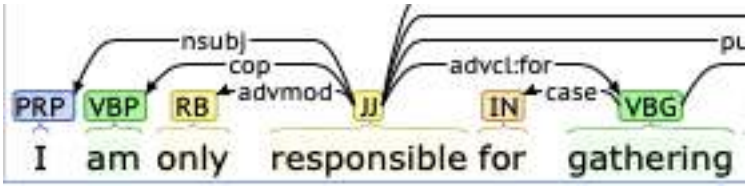
```
"charge": [
    {
        "be": "cop",
        "in": "case",

    }
]
```

5. Add the key “prep” in the dictionary, with the content of an empty list, then added the dep of the real object into the empty list.

```
"charge": [
    {
        "be": "cop",
        "in": "case",
        "prep": [
            "nmod:of"
        ]
    }
],
"responsible": [
```

6. If sometimes an LVC whose information is stored into this file is not recognized, paste the original sentence to the online demo, and check the dep of the object:



For instance, in this sentence, the dep of the real object of the LVC “be responsible for” is “advcl:for” instead of “obl:for”, and the new dep “advcl:for” can be added to the content of the key “prep” in that LVC’s dictionary

```
],
"responsible": [
    {
        "be": "cop",
        "prep": [
            "obl:for",
            "advcl:for"
        ]
    }
]
```